Xebia Internship Day 4

Assignment 1

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Q1) Second Largest Without Duplicates

```
#include <iostream>
#include <vector>
#include <unordered_set>
#include <climits>

using namespace std;

int secondLargest(vector<int>& arr) {
   if (arr.empty()) return -1;

   int largest = INT_MIN, second = INT_MIN;
   unordered_set<int> seen;

   for (int num : arr) {
      if (seen.count(num)) continue;
      seen.insert(num);

      if (num > largest) {
         second = largest;
         largest = num;
      } else if (num > second) {
         second = num;
      }
```

```
    }

    return (second == INT_MIN) ? -1 : second;
}

int main() {
    vector<int> arr1 = {12, 35, 1, 10, 34, 1};
    vector<int> arr2 = {10, 10, 10};
    vector<int> arr3 = {5};
    vector<int> arr4 = {7, 7, 8};

    cout << secondLargest(arr1) << endl;
    cout << secondLargest(arr2) << endl;
    cout << secondLargest(arr3) << endl;
    cout << secondLargest(arr4) << endl;

    return 0;
}
```

Q2) Rotate array K steps to the right

```
#include <iostream>
#include <vector>
using namespace std;

void rotateArray(vector<int>& nums, int k) {
    int n = nums.size();
    if (n == 0) return;
```

```
    k = k % n;
    vector<int> temp(n);

    for (int i = 0; i < n; i++) {
        temp[(i + k) % n] = nums[i];
    }

    nums = temp;
}

int main() {
    vector<int> arr1 = {1, 2, 3, 4, 5, 6, 7};
    int k1 = 3;
    rotateArray(arr1, k1);
    for (int num : arr1) cout << num << " ";
    cout << endl;

    vector<int> arr2 = {-1, -100, 3, 99};
    int k2 = 2;
    rotateArray(arr2, k2);
    for (int num : arr2) cout << num << " ";
    cout << endl;

    return 0;
}
```

Q3) Valid Anagram Check

```
#include <iostream>
```

```
#include <string>
#include <vector>
#include <cctype>
using namespace std;

bool isAnagram(string s1, string s2) {
    vector<int> freq(26, 0);

    for (char c : s1) {
        if (!isspace(c)) {
            freq[tolower(c) - 'a']++;
        }
    }

    for (char c : s2) {
        if (!isspace(c)) {
            freq[tolower(c) - 'a']--;
        }
    }

    for (int count : freq) {
        if (count != 0) return false;
    }
    return true;
}

int main() {
    cout << boolalpha;
```

```
    cout << isAnagram("Listen", "Silent") << endl;
    cout << isAnagram("hello world", "world hello") << endl;
    cout << isAnagram("rat", "car") << endl;

    return 0;
}
```

Q4) Longest Substring Without Repeating Characters

```
#include <iostream>
#include <string>
#include <vector>
using namespace std;

int lengthOfLongestSubstring(string s) {
    vector<int> lastIndex(256, -1);
    int maxLen = 0;
    int start = 0;

    for (int i = 0; i < s.length(); i++) {
        char c = s[i];

        if (lastIndex[c] >= start) {
            start = lastIndex[c] + 1;
        }

        lastIndex[c] = i;
        maxLen = max(maxLen, i - start + 1);
```

```
    }

    return maxLen;
}

int main() {
    cout << lengthOfLongestSubstring("abcabcbb") << endl;
    cout << lengthOfLongestSubstring("bbbbb") << endl;
    cout << lengthOfLongestSubstring("pwwkew") << endl;
    return 0;
}
```

Q5) Top Earning Employees per Department

```
SELECT d.dept_name, e.name, e.salary
FROM Employees e
JOIN Departments d
  ON e.dept_id = d.dept_id
WHERE e.salary = (
    SELECT MAX(salary)
    FROM Employees
    WHERE dept_id = e.dept_id
);
```

Q6) Monthly Order Revenue Report

```
SELECT
    MONTH(order_date) AS month_number,
    SUM(amount) AS total_revenue,
```

```
    COUNT(*) AS total_orders
FROM Orders
WHERE YEAR(order_date) = 2024
GROUP BY MONTH(order_date)
ORDER BY total_revenue DESC;
```

Q7) NA

Q8) NA